William Gao

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EDUCATION

Carnegie Mellon University, College of Engineering **Pittsburgh, PA**

B.S. in Mechanical Engineering with Minor in Robotics: GPA: 3.70/4.0 *Aug 2024 - May 2028*

Relevant coursework: Intro to Electrical and Computer Engineering, Thermodynamics, Mechanics I: 2D Design, Mechanics II: 3D Design, Differential Equations, Fluid Mechanics, Principles of Imperative Computation

In-progress coursework: Heat Transfer, Dynamics, Numerical Methods, Mechanical Design: Methods and Applications

CAD/Design: SolidWorks, 3D Printing (Bambu Studio), Laser Cutting, ANSYS Workbench, Hilti PROFIS, FEA

Programming: Python, Java, MATLAB, C

Hardware/Embedded: Raspberry Pi, Arduino, YOLO Object Detection

PLC/Automation: Do-More Designer, Ladder Logic, FANUC CRX-30iA Cobot Programming, RFID/Barcode/OCR Integration

Fabrication: Crimping, Manual Milling, Soldering, Industrial Wiring from Electrical Drawings

WORK EXPERIENCE

Sepro America **Warrendale, PA**

Engineering Intern *May 2026 - Aug 2026*

- Engineered PLC-based tote inspection sequence (RFID, barcode, OCR verification across dual sides) for a 16-station production scale-out, accounting for ~15 distinct fail conditions and reducing cycle time from 40-60s to under 25s
- Designed and validated a new FANUC robot riser attachment for a new base configuration, achieving a minimum factor of safety of 4-5 via structural FEA in Hilti PROFIS Engineering
- Authored complete FANUC CRX-30iA cobot palletizing algorithm accounting for X/Y/Z moment shifts across single- and dual-tote two-part gripper configurations, dynamic stack-height placement, and dual 0°/90° tote orientations
- Fabricated and wired 20+ industrial cables via crimping and soldering, and wired the robot control system by following electrical drawings, applying a fundamental understanding of relay, transformer, and AC/DC power concepts, saving the company $10,000+ in outsourced labor

Carnegie Mellon University **Pittsburgh, PA**

Teaching Assistant - Physics I for Engineering Students *Aug* 2025 - Dec 2025

- Supported 20+ students by offering 1:1 support and grading homework and exams
- Taught students concepts in kinematics, thermodynamics, rotational dynamics, work, power, and momentum collisions

PROJECTS

Phone-Detection Deterrent Device - Hackberry Pi Hackathon, Carnegie Mellon University *Spring 2026*

- Built a Raspberry Pi device that uses YOLO object detection to recognize phone use and trigger a marble-drop deterrent
- Designed the marble drop chute mechanism and camera stand, wired the system, and contributed to the detection code

Topology-Optimized Structural Bracket Design - Carnegie Mellon University *Fall 2025*

- Iteratively designed and weight-optimized an 8-inch structural bracket in SolidWorks under fixed and roller support constraints, focusing on aggressive material reduction while maintaining load capacity
- Validated designs using ANSYS Discovery FEA (von Mises stress, factor of safety 1-2) and physical laser-cut acrylic testing, converging on a sub-10-gram bracket that successfully supported a 40-lb load

Collaborative Mechanical Design and Fabrication - Carnegie Mellon University *Fall 2024*

- Collaborated with a 4-person team to design and fabricate a fully functional marble dispensing mechanism using SolidWorks, laser-cut wood components, 3D-printed components, and a DC motor drive system
- Performed stress and motion simulations in SolidWorks/ANSYS Discovery to validate mechanical stability and optimize part geometry, ensuring smooth and consistent dispensing operation

LEADERSHIP & ACTIVITIES

Triathlon - Carnegie Mellon University *Jun 2026 - Present*

- Co-founded CMU Triathlon Club, building a community for students new to the sport through group workouts, transition practice, and race day coaching
- Trained and successfully completed an Ironman 70.3 Triathlon (Jun 2026)

Varsity Swim Team - Carnegie Mellon University *Aug 2024 - Aug 2025*

- Ranked as a top 7 fastest swimmer in program history across 4 events; received CSCAA Scholar All-America award